9.8.1.Moments about the origin

Definition:

If X is a random variable and for each positive integer r ($r = 1, 2, ...$) the rth raw moment about the origin '0' is $\mu'_r = E(X^r) = \sum x_i^{\,r}\ p_i$

First order raw Moment

Put $r = 1$, in the definition

$$\mu'_1 = E(X) = \sum x_i\, p_i = \overline{X}$$

This is called the mean of the random variable X. Hence the first order raw moment is mean.

Second order raw moment

Put $r = 2$ then

$$\mu'_2 = E(X^2) = \sum x_i^2 p_i$$

This is called second order raw moment about the origin.

9.8.2. Moments about the mean (Central moments)

Definition:

For each positive integer r, ($r = 1, 2, ...$), the rth order central moment of the random variable X is $\mu_r = E\,(X - \overline{X})^r$

First order central moment:

put $r = 1$ in the definition, then

$$\mu_1 = E\ (X - \overline{X}) = 0 \ \text{(always)}$$

Remark:

The algebraic sum of the deviations about the arithmetic mean is always 0

Second order central moment;

Put $r = 2$ in definition

$$\mu_2 = E\,(X - \overline{X})^2$$

$$= E\,(X^2 - 2\ X \times \overline{X} + \overline{X}^2\ )$$